Question 1

The following code was written by a developer in order to display for ever interleaved (mixed) messages of both Thread T1 and Thread T2. However, the code just displays the message Thread T1 for ever.

Which of the following changes should be made to fix the code to run as intended (i.e. to display mixed messages of Thread T1 and Thread T2)?

```
class T1 extends Thread {
    public void run() {       // line 2
        while (true)
            System.out.println("Thread T1");
    }
}


class T2 extends Thread {
    public void run() {       // line 10
        while (true)
            System.out.println("Thread T2");
    }
}

class ThreadsQ15 {
    public static void main(String[] args) {
        T1 t1 = new T1();     // line 18
        T2 t2 = new T2();     // line 19
        t1.run();             // line 20
        t2.start();           // line 21
    }
}
```

(A) Swap the order of lines 19 and 20

(B) Replace both lines 2 and 10 with: public void start()

Feedback

Feedback for student

Your instructor hasn't added feedback

```
class T1 extends Thread {
    public void run() {       // line 2
        while (true)
            System.out.println("Thread T1");
    }
}


class T2 extends Thread {
    public void run() {       // line 10
        while (true)
            System.out.println("Thread T2");
    }
}

class ThreadsQ15 {
    public static void main(String[] args) {
        T1 t1 = new T1();     // line 18
        T2 t2 = new T2();     // line 19
        t1.run();             // line 20
        t2.start();           // line 21
    }
}
```

(A) Swap the order of lines 19 and 20

(B) Replace both lines 2 and 10 with: public void start()

(C) Replace line 10 with: public void start()

(D) Replace line 20 with: t1.start();

(E) Replace line 21 with: t2.run();

Feedback

Feedback for student

Your instructor hasn't added feedback

Question 2

Which of the following is NOT a characteristic of object oriented programming?

- (A) Instructional decomposition
- (B) Polymorphism
- (C) Abstraction
- (D) Encapsulation

Question 3

PART B - Programming - 40%

You are expected to use a pen and papers and write the source code on papers and scan/ take photograph using a mobile phone and create a single PDF file (named with UoW ID, Ex: w1678756.pdf) with correct orientation and clarity and attach (See Additional content section at the bottom) and then submit. You are not allowed to use the keyboard and type in any program.

Implement the Java classes (with full details of all fields and methods, including constructors and getter, setter methods) which are required to simulate the following problem:

A casino has 10 Fruit Machines that customers are using to play. A fruit machine is a machine in which a player (customer) inserts a coin (worth of 1 pound) every time to play a single game. Following the insertion of a coin, the machine displays to the customer exactly 3 random fruits next to each

Feedback

Feedback for student

Your instructor hasn't added feedback

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Implement the Java classes (with full details of all fields and methods, including constructors and getter, setter methods) which are required to simulate the following problem:

A casino has 10 Fruit Machines that customers are using to play. A fruit machine is a machine in which a player (customer) inserts a coin (worth of 1 pound) every time to play a single game. Following the insertion of a coin, the machine displays to the customer exactly 3 random fruits next to each other (each fruit is chosen randomly from a total of 21 fruits, see below for the details of the probabilities of how the fruits are chosen). If during a single game the machine displays to the player 2 fruits which are the same, the player is credited with 20 pounds. If during a single game the machine displays to the player all 3 fruits which are the same the player, is credited with 100 pounds in their account.

The customer (player) has a name, address, lives in a city (which might be different than that of the casino), a casino balance (credit) which is the money available to play and has a favorite casino with a specific name. A casino has a location (city) and can accommodate up to 100 customers at the same time.

A Fruit machine holds a number of coins (it is assumed that 1 coin is worth 1 pound). Each fruit machine displays 2 fruits of the same kind with probability 10%, if the credit in the machine (number of coins) is greater than 300. (if the credit is less than that, the fruit will never display 2 fruits which are the same). A fruit machine displays 3 fruits of the same kind with probability 2%, if the credit in the machine is greater than 500 (if the credit is less than that, the fruit will never display 3 fruits which are the same). If the credit in the machine is less than 300 then always the fruit machine displays 3 unique fruits.

As part of your implementation, in addition to the classes described in the scenario above, you should also provide a Simulation class in which a player interacts with a fruit machine and plays 100 games with it. You do not need to include getters and setters for every single field. Assuming that you implement the constructor of a class to initialize the fields, you could just provide getters and setters for 1 or 2 fields only.

Feedback

Feedback for student

Your instructor hasn't added feedback

As part of your implementation, in addition to the classes described in the scenario above, you should also provide a Simulation class in which a player interacts with a fruit machine and plays 100 games with it. You do not need to include getters and setters for every single field. Assuming that you implement the constructor of a class to initialize the fields, you could just provide getters and setters for 1 or 2 fields only.

Your Answer

Question 4

How many constructors can a Java class have?

- (A) zero or one
- (B) any number
- (C) one
- (D) a maximum of ten
- (E) two

Question 5

What is the output of the following program?

Feedback

Feedback for student

Your instructor hasn't added feedback

Question 5

What is the output of the following program?

```
class Q7 {
    int x;
    Q7(int x1) {
        x = x1;
    }

    public static void main(String[] arg) {
        Q7 q1 = new Q7(100);
        Q7 q2 = new Q7(100);

        if (q1.equals(q2))
            System.out.println(true);
        else
            System.out.println(false);

        String str1 = new String("123");
        String str2 = new String("123");

        if (str1.equals(str2))
            System.out.println(true);
        else
            System.out.println(false);
    }
}
```

(A) false
false

Feedback

Feedback for student

Your instructor hasn't added feedback

```
        Q7 q1 = new Q7(100);
        Q7 q2 = new Q7(100);

        if (q1.equals(q2))
            System.out.println(true);
        else
            System.out.println(false);

        String str1 = new String("123");
        String str2 = new String("123");

        if (str1.equals(str2))
            System.out.println(true);
        else
            System.out.println(false);
    }
}
```

(A) false
false

(B) false
true

(C) The program will throw an exception during its execution.

(D) true
true

(E) true
false

Feedback

Feedback for student

Your instructor hasn't added feedback

Question 6

What is the output of the following program?

```
interface F {
    void foo();
}

class Y implements F {
    public void foo() {
        System.out.print("Y ");
    }
}

public class Q6 extends Y {
    public void foo() {
        System.out.print("Q6 ");
    }

    public static void main(String[] args) {
        F f1 = new Q6();
        f1.foo();
        F f2 = new Y();
        f2.foo();
        Y y1 = new Q6();
        y1.foo();
    }
}
```

A Q6 Y Y

B Q6 Y Q6

Feedback

Feedback for student

Your instructor hasn't added feedback

```
class Y implements F {
    public void foo() {
        System.out.print("Y ");
    }
}

public class Q6 extends Y {
    public void foo() {
        System.out.print("Q6 ");
    }

    public static void main(String[] args) {
        F f1 = new Q6();
        f1.foo();
        F f2 = new Y();
        f2.foo();
        Y y1 = new Q6();
        y1.foo();
    }
}
```

(A) Q6 Y Y

(B) Q6 Y Q6

(C) Y Y Q6

(D) Y

(E) Q6

Feedback

Feedback for student

Your instructor hasn't added feedback

Question 7

What is the output of the following segment of code?

```
class M {
    int x;
    static int y = 1;

    M(int x) {
        this.x = x;
        ++y;
    }
}

class StaticQ12 {
    public static void main(String[] args) {
        M m1 = new M(10);
        M m2 = new M(20);
        m2.y++;
        m1.x++;

        System.out.println(m1.x + " " + m2.x + " " + m1.y + " " + m2.y);
    }
}
```

(A) 11 20 4 4

(B) 11 20 3 3

(C) 11 20 2 3

(D) 11 21 2 3

Feedback

Feedback for student

Your instructor hasn't added feedback

```
class M {
    int x;
    static int y = 1;

    M(int x) {
        this.x = x;
        ++y;
    }
}

class StaticQ12 {
    public static void main(String[] args) {
        M m1 = new M(10);
        M m2 = new M(20);
        m2.y++;
        m1.x++;

        System.out.println(m1.x + " " + m2.x + " " + m1.y + " " + m2.y);
    }
}
```

- (A) 11 20 4 4
- (B) 11 20 3 3
- (C) 11 20 2 3
- (D) 11 21 2 3
- (E) 11 11 3 3

Feedback

Feedback for student

Your instructor hasn't added feedback

Question 8

Which of the following statements is true?

(A) The super keyword can be used to call a method in any other class of the program

(B) The super keyword can be used to speed up the execution of the current class.

(C) **The super keyword can be used to call a constructor of the parent class, to call a method of the parent class or to access a field of the parent class.**

(D) The super keyword can be used to give access to "hidden" parts of other classes

(E) The super keyword can be used to call a method of the parent class or to access a field of the parent class. It cannot be used to call a constructor of the parent class.

Question 9

What is the output of the following segment of code?

```
class RefsQ11 {
    public static void main(String[] a) {
        String x = new String("a dog ");
        String y = "a cat ";
        String z = x + y;
        x = z;
        x = y;
```

Feedback

Feedback for student

Your instructor hasn't added feedback

Question 9

What is the output of the following segment of code?

```
class RefsQ11 {
    public static void main(String[] a) {
        String x = new String("a dog ");
        String y = "a cat ";
        String z = x + y;
        x = z;
        x = y;

        System.out.println(z + "having a nice day");
    }
}
```

(A) a dog having a nice day

(B) a dog a cat having a nice day

(C) a cat having a nice day

(D) a dog a dog having a nice day

(E) a cat a dog having a nice day

Feedback

Feedback for student

Your instructor hasn't added feedback

Question 10

What is the output of the following segment of code?

```
class MyNumber {
    int i = 5;
    MyNumber(int i) {
        this.i = i;
    }
}
class Q5 {
    void modify(int i) {
        i++;
    }
    void modify(MyNumber m) {
        m.i++;
    }
    public static void main(String[] args) {
        int i = 4;
        Q5 q5 = new Q5();
        q5.modify(i);
        MyNumber m = new MyNumber(2);
        q5.modify(m);
        System.out.println(i + " " + m.i);
    }
}
```

(A) 3 3

(B) 4 3

Feedback

Feedback for student

Your instructor hasn't added feedback

```
        this.i = i;
    }
}
class Q5 {
    void modify(int i) {
        i++;
    }
    void modify(MyNumber m) {
        m.i++;
    }
    public static void main(String[] args) {
        int i = 4;
        Q5 q5 = new Q5();
        q5.modify(i);
        MyNumber m = new MyNumber(2);
        q5.modify(m);
        System.out.println(i + " " + m.i);
    }
}
```

A 33

B 43

C 45

D 34

E 24

Feedback

Feedback for student

Your instructor hasn't added feedback

Question 11

What is the output of the following segment of code?

```
class Parent {
    Parent() {
        System.out.print("Parent ");
    }
}

class Child extends Parent {
    Child() {
        System.out.print("Child ");
    }
}

class GrandChild extends Child {
    GrandChild() {
        System.out.print("Grandchild ");
    }

    GrandChild(int x) {
    }
}

class Q14 {
    public static void main(String[] args) {
        Parent p = new GrandChild();
        Parent p2 = new GrandChild(5);
        System.out.println();
    }
}
```

(A) Grandchild Child Parent Child Parent

(B) Parent Child Grandchild Parent Child Grandchild

Feedback

Feedback for student

Your instructor hasn't added feedback

```
class Child extends Parent {
    Child() {
        System.out.print("Child ");
    }
}

class GrandChild extends Child {
    GrandChild() {
        System.out.print("Grandchild ");
    }

    GrandChild(int x) {
    }
}

class Q14 {
    public static void main(String[] args) {
        Parent p = new GrandChild();
        Parent p2 = new GrandChild(5);
        System.out.println();
    }
}
```

(A) Grandchild Child Parent Child Parent

(B) Parent Child Grandchild Parent Child Grandchild

(C) Grandchild Child Parent Grandchild Child Parent

(D) Parent Child Grandchild Parent Child

(E) Parent Parent

Feedback

Feedback for student

Your instructor hasn't added feedback

Feedback
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Question 12

Which lines (line numbers are shown in comments) of the following program will issue a compiler error? The program consists of three separate files, AccessQ13.java, A.java and B.java.

File AccessQ13.java contains:

```
package f1;

class AccessQ13 {
    public static void main(String[] args) {
        A a1 = new A();
        a1.i = 6;  // line 6
        a1.j = 7;  // line 7
        a1.k = 8;  // line 8
    }
}
```

File A.java contains:

```
package f1;

public class A {
    int i;
    protected int j;
    private int k;
}
```

File B.java contains:

```
package p2;

import f1.A;

class B extends A {
    void foo() {
        i = 5;   // line 15
        j = 10;  // line 16
        k = 20;  // line 17
    }
}
```

(A) lines 8, 15, 16, 17

Feedback

Feedback for student

Your instructor hasn't added feedback

Feedback
learning.westminster.ac.uk/ultra/courses/_88615_1/outline/assessment/_3723223_1/overview/attempt/_4122784_1/review/inline-feedback?attemptId=_4122784_1&mode=inline&c...
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```
    public static void main(String[] args) {
        A a1 = new A();
        a1.i = 6;  // line 6
        a1.j = 7;  // line 7
        a1.k = 8;  // line 8
    }
}
```

File A.java contains:

```
package f1;

public class A {
    int i;
    protected int j;
    private int k;
}
```

File B.java contains:

```
package p2;

import f1.A;

class B extends A {
    void foo() {
        i = 5;   // line 15
        j = 10;  // line 16
        k = 20;  // line 17
    }
}
```

(A) lines 8, 15, 16, 17

(B) lines 8, 16 and 17

(C) lines 8, 15 and 17

(D) lines 8 and 17

(E) lines 15 and 17

Feedback

Feedback for student

Your instructor hasn't added feedback

Feedback
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Question 13

What is the output of the following code?

```
class D {
    private int x;
    private int y = 5;

    public static void main(String[] args) {
        D a1 = new D();
        D a2 = new D();
        D a3 = new D();
        a1.x = a3.y;
        a3 = a1;
        a1 = a3;
        a2.x = 12;
        a3.x = a3.x + 2;
        a1.x = 30;
        a1.y = 21;
        a2.y = 22;
        System.out.println(a1.x + " " + a2.x + " " + a3.x + " " + a1.y +
                           " " + a2.y);
    }
}
```

(A) 30 30 22 22 21

(B) 30 30 30 21 22

(C) 30 30 1 22 22

Feedback

Feedback for student

Your instructor hasn't added feedback

```
class D {
    private int x;
    private int y = 5;

    public static void main(String[] args) {
        D a1 = new D();
        D a2 = new D();
        D a3 = new D();
        a1.x = a3.y;
        a3 = a1;
        a1 = a3;
        a2.x = 12;
        a3.x = a3.x + 2;
        a1.x = 30;
        a1.y = 21;
        a2.y = 22;
        System.out.println(a1.x + " " + a2.x + " " + a3.x + " " + a1.y +
                           " " + a2.y);
    }
}
```

(A) 30 30 22 22 21

(B) 30 30 30 21 22

(C) 30 30 1 22 22

(D) 30 12 30 21 22

(E) 22 30 1 11 22

Feedback

Feedback for student

Your instructor hasn't added feedback

(E) 22 30 1 11 22

Question 14

Which of the following statements is the most accurate about the program below?

```
class A {
    void foo() {}
}

class B extends A {
    void foo() {}
    void foo(int m) {}
}
```

(A) method foo is only overridden in B

(B) method foo is only overloaded in B.

(C) method foo is overridden in A

(D) method foo is overridden in B and also overloaded in B

(E) method foo is overloaded in A.

Feedback

Feedback for student

Your instructor hasn't added feedback

Question 15

What is the output of the following code?

```
class References {
    int x = 8;
    int y=1;

    public static void main(String[] args) {
        References r1 = new References();
        r1.y = 10;
        References r2 = new References();
        r2.y = ++r1.x;

        r1 = r2;
        ++r2.y;

        References r3 = new References();
        r2 = r3;
        System.out.println(r1.x + " " + r1.y + " " + r2.x + " " + r2.y);
    }
}
```

A 8 10 8 1

B 8 1 8 1

C 9 10 8 1

D 10 8 8 10

Feedback

Feedback for student

Your instructor hasn't added feedback

```
class References {
    int x = 8;
    int y=1;

    public static void main(String[] args) {
        References r1 = new References();
        r1.y = 10;
        References r2 = new References();
        r2.y = ++r1.x;

        r1 = r2;
        ++r2.y;

        References r3 = new References();
        r2 = r3;
        System.out.println(r1.x + " " + r1.y + " " + r2.x + " " + r2.y);
    }
}
```

A 8 10 8 1

B 8 1 8 1

C 9 10 8 1

D 10 8 8 10

E 8 10 8 10

Feedback

Feedback for student

Your instructor hasn't added feedback

Question 16

What is the output of the following segment of code?

```
class A2 {
    public static void main(String args[]) {
        String string = "STRING";
        if ("STRING".equals("STRING"))
            System.out.print("Equal ");
        else
            System.out.print("Not Equal ");

        if (string == "STRING")
            System.out.print("Equal ");
        else
            System.out.print("Not Equal ");

    }
}
```

(A) Not Equal Equal

(B) Not Equal Not Equal

(C) Equal Not Equal

(D) Equal Equal

Submission Content

Feedback

Feedback for student

Your instructor hasn't added feedback